

Quantification: 682_z08

=====

PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
-----	-----
lacuna count	10
interior lacuna count	8
frame-edge lacuna count	2
gap bridges added (detection)	24

Interior lacunae only: mean, median and sample SD (n = 8)

measure	kind	mean	median	sd	units
-----	-----	-----	-----	-----	-----
area_px2	lacuna shape	2599.88	2671.5	1127.64	px^2
major_axis_length_px	lacuna shape	103.523	92.8874	29.9285	px
minor_axis_length_px	lacuna shape	33.6052	36.6597	6.7292	px
aspect_ratio	lacuna shape	3.087	2.9214	0.6438	unitless
eccentricity	lacuna shape	0.9401	0.9395	0.0221	unitless
solidity	lacuna shape	0.8296	0.873	0.1073	unitless
roots_count	headline	10	9	4.4078	count
ring_length_r30_px	headline	396.75	356.5	128.332	px
ring_length_r60_px	reported	982	914	329.724	px
owned_length_px	ownership-dependent	1860.56	997.753	1996.48	px
edge_count	ownership-dependent	76	46	85.157	count
mean_edge_length_px	ownership-dependent	25.4567	25.0725	4.7151	px
perimeter_px	normalised	275.426	240.43	89.7432	px
ring_area_r30_px2	normalised	10379.6	9533.5	2334.04	px^2
ring_area_r60_px2	normalised	24427.5	24429.5	5136.87	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9949	1	0.0144	unitless
ring_density_r30	normalised	0.0380064	0.0376643	0.00650669	px^-1
ring_density_r60	normalised	0.0400478	0.0406532	0.00850963	px^-1
roots_per_100px_perimeter	normalised	3.538	3.6845	0.6393	per 100 px
ring_attached_length_r30_px	network v2	317.125	284	135.042	px
ring_attached_length_r60_px	network v2	673.25	584.5	354.146	px
ring_length_w_r30_px	network v2	445.92	397.369	141.602	px
ring_length_w_r60_px	network v2	1103.96	1012.99	365.403	px
sholl_crossings_r10	network v2	10.75	9.5	5.0356	count
sholl_crossings_r20	network v2	14.25	12.5	4.4641	count
sholl_crossings_r30	network v2	16	14	5.1547	count
ring_width_mean_r30_px	width	5.5385	5.518	0.3651	px

Per field

measure	value	units
-----	-----	-----
analysed_area_px2	1.01791e+06	px^2
total_skeleton_length_px	35267	px
canalicular_length_density_per_px	0.0346465	px^-1
junction_count	480	count
junction_density_per_px2	0.00047155	px^-2
lacuna_count	10	count
lacunae_per_px2	9.82e-06	px^-2

skeleton_component_count	720	count
mean_component_length_px	48.9819	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0386951	px^-1
field_density_without_flagged_per_px	0.035027	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	5.7755	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	8	px
width_pixels_used	32998	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	9321	210.577	69.7066	3.0209	0.9436	0.684	1.3112	31.8256	428.162	True
2	2563	90.1026	36.7546	2.4515	0.913	0.9144	1.5381	342.687	87.595	False
3	3496	113.895	40.712	2.7976	0.9339	0.8552	-1.2503	434.254	701.917	False
4	4123	159.751	36.5648	4.369	0.9735	0.7865	1.4743	433.966	210.1	False
5	1521	89.609	30.1271	2.9744	0.9418	0.641	-1.2077	520.501	556.951	False
6	862	61.0771	21.2942	2.8683	0.9373	0.7077	-1.4443	545.484	438.164	False
7	547	50.4784	17.0874	2.9541	0.941	0.8116	-0.4973	672.987	1013.45	True
8	3601	127.976	37.0473	3.4544	0.9572	0.8909	1.4857	713.926	152.006	False
9	1853	92.0913	27.1748	3.3889	0.9555	0.9155	-1.2655	723.731	828.852	False
10	2780	93.6835	39.1668	2.3919	0.9084	0.926	-1.1692	908.613	727.092	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	10	368	955	480.706	21	22.8907	6.2657
2	10	372	992	1236.15	53	23.3235	6.0017
3	13	373	988	966.377	31	31.1734	5.2647
4	18	652	1616	6469.8	277	23.3567	5.9855
5	8	328	683	546.416	23	23.7572	5.7484
6	5	341	840	940.027	56	16.7862	5.1878
7	3	150	415	294.12	8	36.765	4.3724
8	13	526	1305	2897.37	103	28.1298	5.7045
9	7	323	758	1029.13	39	26.3879	5.3314
10	6	259	674	799.222	26	30.7393	5.0843

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	769.541	14367	29344	0.6579	0.6229	0.0256143	0.032545	1.2995
2	234.894	9374	20769	1	0.9593	0.0396842	0.0477635	4.2572

3	313.948	11274	26474	1	1	0.033085	0.0373196	4.1408
4	456.961	14983	32605	1	1	0.043516	0.049563	3.9391
5	233.238	9202	19621	1	1	0.0356444	0.0348096	3.43
6	166.125	7387	17424	1	1	0.0461622	0.0482094	3.0098
7	122.604	3917	10590	0.6212	0.5802	0.0382946	0.0391879	2.4469
8	329.522	12020	29668	1	1	0.0437604	0.0439868	3.9451
9	222.752	9104	23858	1	1	0.0354789	0.0317713	3.1425
10	245.965	9693	25001	1	1	0.0267203	0.0269589	2.4394

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	254	366	398.706	1026.41	12	11	10
2	303	656	421.777	1099.54	11	14	17
3	312	591	420.375	1118.09	13	12	13
4	622	1480	726.186	1818.18	21	23	26
5	227	393	374.363	776.767	5	12	12
6	265	578	373.522	926.441	7	11	13
7	88	194	170.924	478.63	4	6	7
8	375	815	589.902	1448.74	13	19	21
9	221	447	368.647	884.46	8	13	15
10	212	426	292.593	759.453	8	10	11

